Smart Power Switch-1

(*SPS-1*)

BETA

Software: V1.1

User Manual

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Contents

Introduction

Introducing the Smart Power Switch (SPS-1), a cutting-edge device designed to revolutionize the way you manage and protect your electrical equipment. The SPS-1 safeguards your 12 volt equipment from electrical anomalies, ensuring their longevity and optimal performance.

Vehicle manufacturers are transitioning away from mechanical fuses or circuit breakers and moving to solid state circuit protection. The SPS-1 provides the same protection in a stand alone product.

The SPS-1 offers robust, solid-state protection features that are essential for modern, transceivers, station accessories, and other electronics. It provides **over current** protection, acting like a fuse or circuit breaker to prevent damage from excessive current. Additionally, it offers **over voltage** protection to shield your equipment from potentially harmful voltage spikes, and **under voltage** protection to prevent erratic or unexpected operation due to insufficient voltage.

The SPS-1 is versatile enough to support multiple use cases. It can operate in three distinct modes to suit various needs:

- **Stand Alone** operation: Utilize a simple, low current switch and LED indicators to control power and provide status indication.
- **Remote (DCN)** operation: Support complete power control status updates and telemetry as part of a Device Control Network, or through a straightforward RS-485 serial interface.
- **Mobile Mode**: Enjoy protection, simple switch control, and automatic power off and resume based on vehicle battery voltage.

The SPS-1 contiains no mechanical switching devices. This allows much faster response to voltage or current events with no fear of stuck or welded contacts.

The Smart Power Switch (SPS-1) is a comprehensive solution to bring peace of mind to your station operation.

Features

- Capable of switching more than 30 Amps of operating current
- Overcurrent protection configurable from 5 amps to 25 amps nominal fuse equivalent
- Overcurrent trip operates like a fuse by allowing brief inrush currents, but tripping immediately on a hard short-circuit
- Overcurrent trips can be configured for manual or automatic reset
- Overvoltage protection configurable to user selected voltage (default set to 14.5V)
- Undervoltage protection configurable to user selected voltage (default set to 10.5V)
- Local, hardwired power control with a simple low-current switch closure
- Built in status and fault LED status indicators w/ support for driving external LED indicators
- Standard 45 Amp Anderson Power-Pole connections
- Remote operation via a simple RS-485 serial interface
  (Fully compatible with *Sierra Radio Syste*ms *<u>Device Control Network</u>* protocol)

Operation

Protection

The protection features of the SPS-1 are active in every mode of operation to protect the radio or other electronics connected to the switch. The primary protections are over-current and over-voltage. Under-voltage protection is also provided. However, this is less critical as low voltage can cause erratic operation, but does not typically cause any damage to electronic equipment.

The SPS-1 uses an Infineon PROFET™ series smart high-side switch. These switches are specifically designed for automotive applications. They include self-protection for over-current and over-temperature conditions. The SPS-1 combines the smart power switch with a microcontroller to monitor voltage and current. This allows trip points to be adjusted to precise levels in order to protect sensitive electronics.

The control program uses interrupt based software to read voltage and current 1000 times/second. The controller also incorporates an independent watchdog monitor that will turn off the switch and reset the controller in the unlikely event of a software malfunction or lockup.

Over-voltage protection is performed in the interrupt routine. An over-voltage condition will turn off the switch within 1 millisecond. Once the over-voltage condition clears, the switch will automatically turn back on after a ½ second delay.

Over-current protection is based on an i2t (i-squared-t) model. This mimics how fuses operate and defines a specific amount of energy required to trip an over-current fault. This allows the switch to pass a small, brief inrush overload without tripping and still trip instantly for a high-current short circuit. The time to trip is based on how severe the overload is.

Over-current faults can be reset manually or automatically at the user's option. Manual reset simply requires turning the switch off and back on again. The default delay before an automatic reset is 5 seconds. The delay time can be changed through the serial interface. If 3 over-current faults occur in rapid succession with automatic reset enabled, the controller will discontinue automatic resets and wait for a manual reset.

Under-voltage monitoring requires that the under-voltage condition persists for ½ second before the switch will be turned off. This allows for brief power supply droops without causing false trips. Similar to over-voltage, the switch will turn back on once the voltage rises above the under-voltage limit for ½ second.

Connections and Controls

Power Connections

The power source is connected to the IN PowerPoles. The power source can be a power supply, battery, multiple sources, or an automotive 12 volt source. Since the SPS-1 includes the fuse function, consider placing it as close as possible to the power source.

NOTE: The SPS-1 is not designed to withstand the engine compartment environment and should NOT be mounted under the hood or in any vehicle compartment exposed to outside weather conditions or directly exposed to engine heat.

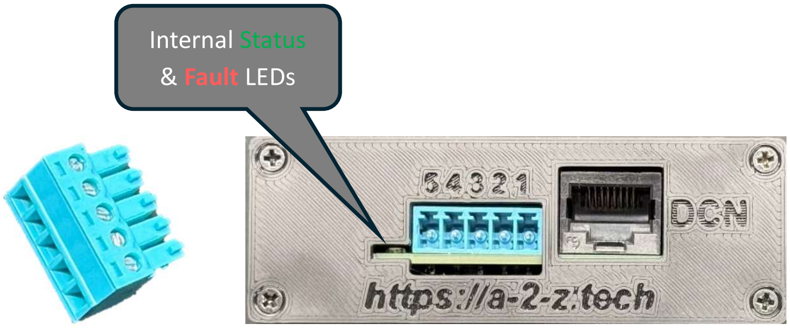
The switched load is connected to the OUT PowerPoles.

If the SPS-1 is mounted some distance from the power source the following must be considered:

- Use a large enough wire gauge to minimize voltage drop based on the expected load current.
- Place a fuse at the power source to protect the wiring from the source to the SPS-1 in the event of a catastrophic wiring or other failure.

Control Connections

The SPS-1 can be controlled locally through the 5 pin terminal block connector, or it can be controlled remotely through the DCN connector. A plug in wire terminal block is included for the terminal block connector. The terminal block connections are shown below:

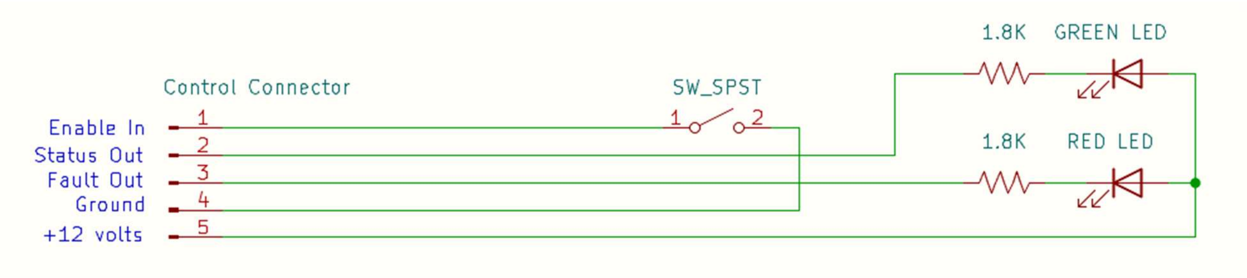


Local Control Connector Pinout

PIN	Name	Description
1	Power Enable	Ground this pin to enable the output.
2	Power Status	This pin will be grounded when the output is ON. (see below for other status conditions)
3	Fault	This pin will be grounded when there is a fault condition. (see below for other indications)
4	Ground	This pin provides grounding point for the enable signal.
5	12V	This pin provides a connection to the input power pin for use with external status indicators (50ma Max).

The Power Status and Fault Pins can be used to connect external status LED indicators. It is suggested to use Green for status and Red for fault to match the internal LED indicators.

Example local control wiring



LED and Status/Fault pin indications

Power Status	FAULT	DESCRIPTION
OFF	OFF	Power output is OFF. Switch is in Idle state.
ON	OFF	Power output is ON. No Fault Conditions exist.
OFF	ON	Power output is OFF. An overcurrent trip has occurred.
Slow Flashing (“Winking”)	ON	Power output is OFF. Undervoltage condition exists.
Rapid Flashing	ON	Power output is OFF. Overvoltage condition exists.
Slow Flashing (“Winking”)	OFF	Power output is OFF. SPS-1 is in Mobile Mode. Low voltage (default 12.8V) timer has expired. Switch is in Standby, waiting for voltage to go above upper voltage setpoint (default 13.3V).

Switch Settings

Common control settings and options can be selected with 2 rotary switches on the SPS-1 circuit board. To access the switches, remove the 4 screws from either end plate on the case and slide the circuit board out of the case. The 2 control switches are the SET switch and the DCN ADDR switch.

Set Switch

The SET switch selects common over-current and operating mode options. The DCN ADDR switch selects the device address for this switch when connected to a Device Control Network (See Serial Port (DCN) Operation below).

SET SWITCH OPTIONS

POSITION	SETTING
0	Settings taken from EEPROM (Allows changes to delays, retries, Mobile Mode options)
1	5 amp fuse – Manual Reset
2	10 amp fuse – Manual Reset
3	15 amp fuse – Manual Reset
4	20 amp fuse – Manual Reset
5	25 amp fuse – Manual Reset
6	5 amp fuse – Auto Reset – 5 sec delay
7	10 amp fuse – Auto Reset – 5 sec delay
8	15 amp fuse – Auto Reset – 5 sec delay
9	20 amp fuse – Auto Reset – 5 sec delay
A	25 amp fuse – Auto Reset – 5 sec delay
B	5 amp fuse – Mobile Mode
C	10 amp fuse – Mobile Mode
D	15 amp fuse – Mobile Mode
E	20 amp fuse – Mobile Mode
F	25 amp fuse – Mobile Mode

NOTE: See DCN Commands for details on setting EEPROM options

DCN ADDR

The DCN ADDR switch sets the device address for this switch when connected to a DCN or compatible RS-485 serial network via the DCN connector.

Switch positions 1 – F set the device address to 01 through 0F Hex, as indicated on the switch. In position 0, the SPS-1 will read it's address from EEPROM and will use that instead of the address indicated on the switch. Address 00 is reserved for the primary controller and is not a valid address for remote devices.

See Serial Port (DCN) Operation for more details.

Mobile Mode Operation

Mobile Mode is intended for use when powering equipment in a vehicle. When connecting radio gear in a vehicle, it is often desirable to have the power cable for the radio wired directly to the vehicle battery. It is often also desirable to have power removed from the radio equipment when the vehicle is off. This requires wiring a relay into the power wires and also finding a suitable ignition controlled 12 volt source to connect to the relay coil.

With Mobile Mode, the SPS-1 can be connected between the power cable and the radio and no additional ignition controlled source is necessary. A remote switch and status LEDs can optionally be wired in (as shown in Control Connections) and located near the driver's position, if desired.

In Mobile Mode, the SPS-1 will monitor the input voltage. When the enable switch is on, the load will remain powered as long as the input voltage is above 12.8 volts. When the voltage drops below 12.8 volts, a timeout-timer is started. The radio will remain powered until the timer expires. (The default timeout period is 10 minutes). When the timer expires, the output will turn off and the SPS-1 will enter Stand-by mode. This mode is indicated by the Power Status LED "winking" at 1 second intervals.

When the input voltage goes above 13.3 volts, the power output will be re-enabled and the switch will continue with normal operation. If the input voltage goes above 13.3 volts during the timeout period, the timeout timer will be cancelled and normal operation will continue.

While the switch is in stand-by mode, the output can be re-enabled by toggling the enable input signal off and back on. This will turn-on the output. If the input voltage remains below 12.8 volts, the output will turn off again after another timeout period.

The lower and upper trigger voltages and the timeout period can all be modified through the DCN serial port.

In stand-by mode, the SPS-1 requires approximately 7ma to operate. A typical car battery capacity is 60 to 80 amp-hours. Therefore the SPS-1 could operate in stand-by mode in a vehicle for months before substantially affecting the vehicle battery's state-of-charge.

Serial Port (DCN) Operation

The SPS1 supports multi-drop serial port operation over an RS-485 network. It conforms to the DCN message protocol defined by Sierra Radio Systems.

You can learn more about the Device Control Network and Station controllers on the SRS website here:
https://www.packtenna.com/station-controller.html#/

Or on their Groups.io group here:
https://srs.groups.io/g/station-controllers

RS-485 communication is very robust allowing communications over very long distances, up to 1000 feet or more. It is a simple 2-wire protocol. Communication is half-duplex (one way at a time) so multiple devices can be connected and communicate over the same wire pair.

The DCN definition also provides for 12 volt power to be available on the RJ-45 network jack. **The SPS1 controller can be powered by this network power (if available).** This allows the controller to continue to operate and respond to DCN commands and queries if the primary switch input power is lost.

Also, it is important to note that all DCN commands and responses consist of standard ASCII text. So you can use a simple terminal program on Windows, MAC, or Linux to send commands and get responses from your SPS-1

DCN RJ-45 Connector Wiring

1 – RS-485 A
2 – RS-485 B
3 – NC
4 -- +12 VDC
5 -- +12 VDC
6 – NC
7 – GROUND
8 – GROUND

NOTE: This wiring DOES NOT conform to standard Ethernet wiring. However, a straight-through Ethernet jumper cable can be used to connect 2 DCN compatible devices together.

DCN Message format

DCN messages use a human-readable, simple ASCII format. This allows messages to be easily sent and read using a terminal program, such as PuTTY or similar.

General Message format:

/FFTT:CMD,P1,P2,P3<etc>:XX<cr>

Where:

- '/' is required as the first character of the message
- FF is the address of the device sending the message. This can be any 2 characters, but is typically a 2 digit number. By convention, the master controller has an address of 00.
- TT is the address of the device that the message is directed to. It is the same format as the 'from' address.
- ':' is required to separate the addresses from the main message content.
- CMD is the command or message type string. (See supported commands below)
- P1, P2, P3, etc. are the message parameters. These are specific to each command or message type. Parameters are separated by commas.
  The maximum number of parameters supported varies by device.

  NOTE: The Command and Parameters combined are the message Payload and fall between the 2 ':' characters.
- ':' is a required separator character.
- 'XX' is a future CRC error check. Currently, it is the literal characters 'XX'
- <cr> - Carriage Return terminates the command string.

Special broadcast message format:

//CMD,P1<etc><cr>

- '//' - 2 consecutive start characters indicate a broadcast message. No From or To addresses are included. All devices on the network will respond to this message.
- CMD,P1 – Command and parameters follow the same format as described above.
- <cr> - The message is terminated with a Carriage Return. There is no terminating separator nor CRC characters.

The broadcast message format can be used to send a command to all devices. It is also typically used for programming or configuring a device that is directly connected to a computer with no other devices connected.

SPS-1 DCN Commands

Summary of Commands

COMMAND	Parameters	Description
PWR	0 or 1	Power output Enable
STATE	None	Returns the current operating state in UPDATE message
UVSET	Millivolts	Sets the under-voltage trip level in millivolts
OVSET	Millivolts	Sets the over-voltage trip level in millivolts
OCSET	Milliamps, Reset type, Delay	Sets over-current trip level in milliamps and over-current behavior.
MOBSET	Enable, Off-Millivolts On-Millivolts Timeout min.	Sets Mobile Mode operation parameters.
CONFIG	None	Returns current configuration settings in SETTINGS message
VERSION	None	Returns current firmware version in FWVER message (V1.1)
CALLO	Milliamps	Calibration Low current setting **IMPORTANT:** Read calibration section before using this command.
CALHI	Milliamps	Calibration High current setting **IMPORTANT:** Read calibration section before using this command.
CALSET	Scale Setting Offset Setting	Updates the stored calibration setting to the values provided. Can be used to restore calibration parameters if EEPROM data is cleared or becomes corrupted. (V1.1)
LOGS	None	Returns SPS-1 log data
SETADDR	2 character DCN address	Sets device DCN address
RSTLOGS	None	Resets log data (except total on time)
REBOOT	None	Forces a processor restart – equivalent to removing and restoring power. (V1.1)

NOTE: The SPS-1 ignores invalid commands or any message that is not formatted correctly. There is no response or direct acknowledgement of operational commands. The execution of PWR and the various SET commands can be confirmed by checking the response to STATE, CONFIG, or LOGS commands.

Command Details

- **PWR**,n
  n=1 requests output power ON
  n=0 requests output power OFF
  Power will only be turned ON if the input voltage is within Undervoltage & Overvoltage limits.
  This command works in conjunction with the local, hardwired Enable signal. If either the local enable or the DCN Enable is on (1), then power will be enabled. BOTH enable signals must be OFF to turn power off.

- **STATE** <no parameters>
  Returns the current state of the SPS1 in the following format:

  /fftt:UPDATE,SPS1,D,L,P,FS,r,VV.VVV,AA.AAA,WD:XX<cr>
  Where:
  - ff = The address of this device
  - tt = The address of the device that sent the STATE command (or 00 if State was sent in a broadcast message)
  - UPDATE = Response message identifier
  - SPS1 = The name of this device
  - D = 1 If DCN output request is ON, 0 if DCN output request is OFF
  - L = 1 if Local Enable input is ON, 0 if Local Enable input is OFF
  - P = 1 if output power switch is ON, 0 if output power switch is OFF
  - FS = Fault Status – " "- No Fault, "UV"- Undervoltage Fault, "OV"- Overvoltage Fault "OC"- Overcurrent Fault
  - r = The number of OverCurrent reset retries that have been attempted.
    (NOTE: This only applies if OverCurrent Auto Reset is set to TRUE, otherwise this value will be zero)
  - VV.VVV – The voltage on the power input terminals
  - AA.AAA – The current in amps flowing through the switch
  - WatchDog status -
    This field should normally be "--". If this field contains WD, then the internal controller has experienced a WatchDog reset. The FAULT LED will flash rapidly and the Output will be turned off if this occurs. The DCN and Local Enable signals must be turned Off to reset the switch. Then the switch will operate normally again. The FAULT LED flashing will be reset the next time the Output is turned on.
    "WD" will remain in this field until power is removed from the SPS-1 or until a REBOOT command is executed.
    NOTE: Please notify A2Z Tech if this condition occurs.

- **UVSET**,nnnnn
  Sets Undervoltage trip limit
  nnnnn= trip limit in **<u>millivolts</u>**.
  This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.
  Value must be between 6000 and 18000. Any other values or non-numeric characters are invalid and the command will be ignored.
  The result of this command is written to EEPROM and persists through power cycles. The new under-voltage limit takes effect immediately.

- **OVSET**,nnnnn
  Sets Overvoltage trip limit
  nnnnn= trip limit in **<u>millivolts</u>**.
  This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.
  Value must be between 6000 and 18000. Any other values or non-numeric characters are invalid and the command will be ignored.
  The result of this command is written to EEPROM and persists through power cycles. The new over-voltage limit takes effect immediately.

- **OCSET**,nnnnn,a,t
  Sets Overcurrent trip limit
  nnnnn= trip limit in **<u>milliamps</u>**
  NOTE: The minimum trip current allowed is 5000 and the maximum allowed is 35000.

  a= Automatic Reset selection – 0= Manual Reset 1= Automatic Reset
  t= Reset Delay in seconds (Whole seconds only. No decimals)

  When configured for Manual Reset an overcurrent trip is reset by toggling the enable OFF and then back ON.

  This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.
  The result of this command is written to EEPROM.
  IMPORTANT: The new over-current parameters take effect immediately. However, unless the SET switch is set to the 0 position, the over-current level will be reset based on SET switch position whenever input power to the SPS-1 is removed and restored.

- **MOBSET**,enable,off,on,time
  Mobile Mode Settings

  enable= 0 to Disable Mobile Mode, 1 to Enable Mobile Mode
  off= Off Threshold in mv. If voltage dips below this voltage, the timeout timer starts counting down.
  NOTE: This must be at least 1V (1000mv) greater than the Under Voltage trip level.
  on=On Threshold in mv. When the voltage goes above this level, the output will be turned back on if the timeout has expired.
  If the output is on and timeout timer is counting down, it will be cancelled.
  NOTE: This must be at least 1V (1000mv) less than the Over Voltage trip level.
  time=Timeout Timer period in minutes. This is how long the output will remain on once input voltage dips below the Off Threshold. This value must be at least 1 and no more than 600 minutes.

  This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.
  The result of this command is written to EEPROM.
  IMPORTANT: The new Mobile Mode parameters take effect immediately. However, unless the SET switch is set to the 0 position, Mobile Mode parameters will reset based on the SET switch position whenever input power to the SPS-1 is removed and restored.

- **CONFIG** <no parameters>
  Returns the current configuration of the SPS1 in the following format:

  /fftt:SETTINGS,SPS1,xx,uuuuu,ooooo,ccccc,a,t,m,moff,mon,mto,cal,ofst:XX
  Where:
  - ff= The current DCN Address of this device
  - tt= The address of the device that sent the CONFIG command. (00 if received in a broadcast command)
  - SETTINGS= The message type of this response
  - SPS1= The name of this device
  - xx= The 2 digit hex address currently programmed in EEPROM for this device.
    NOTE: This is not necessarily the current address of the device.
  - uuuuu= Undervoltage trip level in millivolts
  - ooooo= Overvoltage trip level in millivolts
  - ccccc= Overcurrent trip level in milliamps
  - a= Autoreset setting. 0= Manual Reset 1= Automatic Reset
  - t= Autoreset time delay in seconds
  - m= Mobile Mode Status 1=Mobile Mode Enabled, 0=Mobile Mode Disabled
  - moff= Mobile Mode Off Threshold in mv
  - mon= Mobile Mode On Threshold in mv
  - mto= Mobile Mode Timeout time in minutes
  - cal= Calibration scaling for current sensing readout & trip level (See Calibration section)
  - ofst= Calibration offset for current sensing readout & trip level (See Calibration section)

- **VERSION** <no parameters (Version 1.1 and later)
  Returns the current firmware version in the following format:

  /fftt:FWVER,SPS1,<ver>:XX
  Where:
    - ff= The current DCN Address of this device
    - tt= The address of the device that sent the CONFIG command. (00 if received in a broadcast command)
    - SETTINGS= The message type of this response
    - SPS1= The name of this device
    - <ver>= The current firmware version as a string (10 characters max)
-
- **CALLO**,ccccc
  Calibration Low command
  ccccc= measured current in milliamps

  IMPORTANT: See Calibration section for details on using this command.

- **CALHI**,ccccc
  Calibration High command
  ccccc= measured current in milliamps

  IMPORTANT: See Calibration section for details on using this command.

- **CALSET,ss.sss,ooooo (V1.1 firmware and later)**
  ss.sss= Current reading scale factor. This is always a positive number.
  ooooo= Current reading offset in milliamps. This may be positive or negative.

  Calibration Set Command. This sets the SPS-1 current reading calibration to the values sent in the command. The command is provided to reload the calibration data in the event that the SPS-1 internal EEPROM settings get corrupted. The scale and offset values are the same values reported in the SETTINGS response to the CONFIG command.
  You can use the CONFIG command to get these values and record them for safe keeping. You can also contact A2Z Tech and provide your SPS-1 serial number and the factory calibration values will be sent to you.

- **LOGS** <no parameters>
  Returns the fault history of the SPS1 in the following format:

  /fftt:HISTORY,SPS1,,tot,uv,ov,oc:XX<cr>
  Where:
    - ff = The address of this device

- tt = The address of the device that sent the LOGS command (or 00 if State was sent in a broadcast message)
- HISTORY = Response message identifier – Historical Data
- SPS1 = The name of this device
- EMPTY FIELD – Between the SPS1 identifier and the Total Time field, there is an extra comma. This is a known bug and will be corrected in the next release.
  NOTE: This is corrected in Version 1.1
- Tot= The total amount of time the output has been ON in tenths of a second.
  NOTE: This value can be up to 4,294,967,296 tenths (~ 13.5 years)
  This value is NOT reset by the RSTLOGS command.
- uv= The number of Undervoltage trips since production or since data was reset
- ov= The number of Overvoltage trips since production or since data was reset
- uc= The number of Overcurrent trips since production or since data was reset

SEE: RSTLOGS Command

- **SETADDR**,xx
  Set the DCN Address of this device
  xx= 1 or 2 hex digits

  The SPS1 internally stores it's address as a single byte of data. Therefore it supports any one of 255 addresses from 01 through FF. It will not accept an address of 00.

  The result of this command is stored to EEPROM. The current DCN address of this device is also **immediately set to the commanded address**.

  IMPORTANT: In order for this address to persist after power is removed from the SPS1, the ADDR switch must be set to **'0'**, otherwise the DCN Address will revert to the address set on the switch after power is removed and re-applied to the SPS1.

- **RSTLOGS** <no parameters>
  This command resets the stored fault memory of the device to zero for each fault type.
  NOTE: This does not reset the total accumulated ON time value.

- **REBOOT** <no parameters> **(Version 1.1 firmware and later)**
  This command reboots the SPS-1 as if a power cycle had occurred.

Calibration

IMPORTANT: Your SPS-1 unit was calibrated at the time of manufacture. It is not necessary to perform the calibration procedure unless you wish to experiment or attempt to further improve the accuracy of current monitoring around a specific current range or in specific temperature conditions.
The calibration procedure is straightforward, and the calibration calculations are automatically performed by the SPS-1. HOWEVER, **there is no way to restore the factory calibration once you perform your own calibration procedure.**

Version 1.1 firmware: The CALSET command has been added to load predetermined scale and offset values into the device. This allows the original factory values to be restored.

NOTE:

> Calibration can improve the accuracy of the SPS1. However, the SPS1 is not designed as a measurement instrument. It may not reliably report current levels of less than 3-4 amps. The current values reported are not suitable for measuring battery state-of-charge or for any use case requiring high accuracy measurements.

The power switching device used in the SPS1 is smart power switch that includes internal thermal protection, overcurrent protection and current sensing. The switch is designed to support intermittent loads of over 100 amps. Because of this, the scaling factor used for current sensing built into the switch is very large. The SPS1 uses the internal current sensing of the switch both for overcurrent fault trips and for reading out load current from the STATE command.

The current sensing built into the switch can have scaling errors as much as 20% and offset errors amounting to several amps in either direction. The calibration performed during the SPS-1 manufacture allows overcurrent protection to be performed with reasonable accuracy down to levels as low as 5 amps.

If more accurate current protection is desired at lower levels, or if a more accurate load current readout is desired, the SPS1 supports a calibration procedure based on the power switch manufacturer's calibration instructions. The SPS1 will calculate calibrated scaling and offset values based on a simple procedure using the CALLO and CALHI calibration instructions.

After calibration, the SPS1 will typically report load currents within 100ma of the actual value for load currents of 1.5 amps or higher. The offset correction will also prevent erroneously reporting small load currents when no load is present.

Calibration procedure

The calibration procedure is fairly simple. All scaling and offset calculations are automatically performed by the SPS1. However, it is very important that procedure be performed in the proper sequence and using loads that meet the stated requirements. To perform a calibration you must be able to apply 2 different steady-state loads to the switch. You must also have the ability to accurately measure the load current applied.

Calibration is performed in two steps. First with a lower load, and then with a higher load. The loads must meet the following requirements:

- Minimum load for Lower calibration: 3 amps.

- Minimum difference between Higher and Lower loads: 5 amps.
  (The greater the difference, the better. A difference of at least 10-15 amps should be used, if possible.)

The loads need to be steady-state loads. DO NOT use loads with active electronics such as a transceiver or amplifier. Suitable loads would be high-power resistors, incandescent car headlamps (several in parallel are required), a 12 volt heater, or an electronic load instrument designed to produce constant DC loads.

You must be able to accurately measure the current being drawn by your load. This could be done with a multi-meter capable of reading currents of at least 20 amps, an accurate ammeter, or a calibrated current shunt used in conjunction with a multi-meter.

The procedure is as follows:

1. (optional) Issue a CONFIG command and note the Scale and Offset calibration values.
2. Insure that the Overcurrent trip level is set higher than the maximum test load you intend to use.
3. Insure that you have a power supply or battery capable of delivering more than the higher test load.
4. Insure that you have an accurate current measuring device for the load levels you will be applying.
5. Enable the SPS1 Power output. (This can either be via the local enable or DCN command)
6. Connect the 1st (lower current) load.
7. Measure the actual load current.
8. Issue the CALLO,ccccc command where ccccc is the measured current in milliamps.
9. Connect the 2nd (higher current) load.
10. Measure the actual load current.
11. Issue the CALHI,ccccc command where ccccc is the measured current in milliamps.

The CALHI command will calculate scaling and offset values and store them to EEPROM. Your SPS1 is now calibrated. You can issue the CONFIG command and see updated calibration values. You should find that current values reported are within 100 ma of the actual load value.